

STATISTICS — STUDY GUIDE

PART 2

Time Series

Cautions · Autocorrelation · Stationarity · Dynamic Models

Enhanced Edition — Beginner-Friendly with Prerequisite Foundations

How to read this guide

KEY POINT	Key concept or definition — must know.
FORMULA	Equation as it appears on the formula sheet.
INTERPRETATION	How to read/apply the result in plain English.
NEW CONCEPT	Background explanation for readers new to statistics.
SPSS	Menu path and what to click in SPSS.
WATCH OUT	Common exam pitfall or subtle distinction.

PREREQUISITE

Statistics Foundations

Everything you need before reading this guide

If you have never studied statistics before, read this section first. It explains every concept that the rest of the guide takes for granted — regression, hypothesis testing, Greek notation, and more. If you have already taken an introductory statistics course, you can skip ahead to Chapter TS_A.

P.0 — Greek Letters and Mathematical Notation

NEW CONCEPT

Why Greek letters? Statistics uses Greek letters to distinguish population quantities (the true, unknown values) from sample quantities (the estimates we compute from data). Here are the ones used throughout this guide:

Symbol	Name	What it usually means in this guide
α (alpha)	alpha	Constant / intercept in a model; also significance level of a test
β (beta)	beta	Slope coefficient — how much Y changes when X changes by 1
γ (gamma)	gamma	Coefficient on a lagged Y variable in AR models
ε (epsilon)	epsilon	Error term — the part of Y that the model cannot explain
μ (mu)	mu	True (population) mean of a variable
σ (sigma)	sigma	Standard deviation; σ^2 is variance
σ^2 (sigma ²)	sigma squared	Variance — average squared distance from the mean
ρ (rho)	rho	Autocorrelation coefficient between errors one period apart
λ (lambda)	lambda	Coefficient on a time-trend variable
δ (delta)	delta	Slope of the long-run trend in certain AR models
χ^2 (chi-sq.)	chi-squared	A distribution used in hypothesis tests (e.g. LMSC)
Δ (Delta)	Delta	Difference operator: $\Delta X_t = X_t - X_{t-1}$
Σ (Sigma)	Sigma	Summation: $\Sigma \beta_k$ means $\beta_0 + \beta_1 + \dots + \beta_q$

P.1 — What Is Regression?

NEW CONCEPT

The core idea. Regression is a mathematical tool that finds the best-fitting straight line (or curve) through a cloud of data points. We want to explain or predict one variable (Y, called the dependent or response variable) using one or more other variables (X, called independent or explanatory variables).

NEW CONCEPT

Example. Suppose we want to understand how education spending (X, in dollars per child) relates to GDP growth (Y, in %). We collect data for many years and find the line that best fits those data points. That line is our regression model.

The simple (one X) linear regression equation is:

FORMULA

$$Y_t = \beta_0 + \beta_1 X_t + \varepsilon_t$$

β_0 = intercept (value of Y when X = 0)

β_1 = slope (how much Y changes for a 1-unit increase in X)

ε_t = error / residual (the part of Y not explained by X)

NEW CONCEPT

Multiple regression simply adds more X variables: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$. Everything in this guide is a special case of multiple regression applied to data collected over time.

P.2 — What Is OLS (Ordinary Least Squares)?

NEW CONCEPT

OLS is the most common method for estimating regression coefficients. It finds the values of $\beta_0, \beta_1, \dots$ that minimise the sum of squared residuals — i.e. it finds the line that is as close as possible to all data points at once.

NEW CONCEPT

BLUE stands for Best Linear Unbiased Estimator. Under the five classical assumptions (see P.4), OLS is proven to be the best possible way to estimate β — 'best' meaning it has the smallest possible variance among all unbiased linear methods.

P.3 — Residuals, R^2 , and Model Fit

NEW CONCEPT

Residual (e_t). The difference between the actual observed value and the value predicted by the regression line: $e_t = Y_t - \hat{Y}_t$. A good model has residuals that look like random noise — no pattern.

NEW CONCEPT

R^2 (R-squared). A number between 0 and 1 that tells you how much of the variation in Y is explained by your model. $R^2 = 0.80$ means the model explains 80% of the variation. A higher R^2 is generally better, but a high R^2 does not automatically mean the model is correct — especially with time-series data (see spurious regression, Section C.1).

P.4 — The Five Classical Assumptions (A1-A5)

OLS is BLUE only when these five conditions hold. The time-series part of this course focuses on what happens when A3 fails.

NEW CONCEPT

A1: $E(\varepsilon_t) = 0$ — On average the errors are zero. The model is neither systematically too high nor too low.

NEW CONCEPT

A2: $\text{Var}(\varepsilon_t) = \sigma^2$ (constant) — Called homoskedasticity. The spread of errors is the same across all observations. If the spread changes (e.g. errors get bigger over time), we have heteroskedasticity and A2 is violated.

NEW CONCEPT

A3: $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$ — Errors are uncorrelated with each other. This is the assumption most likely to fail in time-series data. If yesterday's error influences today's error, A3 is violated — we call this autocorrelation.

NEW CONCEPT

A4: X variables are non-random and not exact linear combinations of each other. The second part means no perfect multicollinearity — you cannot have one X variable that is simply a multiple of another.

NEW CONCEPT

A5 (optional): ε is normally distributed. Needed for exact t - and F -tests in small samples. With large samples, the Central Limit Theorem makes normality less critical.

P.5 — Hypothesis Testing: H_0 , H_1 , p -values, and Rejection

NEW CONCEPT

The logic of hypothesis testing. We start with a null hypothesis (H_0) — a default assumption we want to challenge. We then compute a test statistic from the data and ask: 'How likely is it to see a result this extreme if H_0 were actually true?'

Term	Plain-English meaning
H_0 (null hypothesis)	The 'boring' baseline — e.g. 'there is no effect', 'the coefficient is zero', 'there is no autocorrelation'
H_1 (alternative hypothesis)	What we suspect might be true if H_0 is wrong — e.g. 'there IS an effect'
Test statistic (t , F , LM , ...)	A number computed from data that measures how far the data are from what H_0 predicts

p-value	Probability of getting a test statistic this extreme if H_0 were true. Small p-value = strong evidence against H_0
Significance level (α)	The threshold we set in advance (commonly 0.05). If p-value < α , we reject H_0
Reject H_0	The data give enough evidence to conclude H_0 is false. Does NOT prove H_1 with certainty
Fail to reject H_0	Not enough evidence against H_0 . Does NOT prove H_0 is true — the test just could not disprove it
Critical value	The threshold value of the test statistic beyond which we reject H_0

WATCH OUT

Rejecting H_0 is not always 'good news.' In the LMSC autocorrelation test, rejecting H_0 means autocorrelation IS present — that is a problem. In the Dickey-Fuller test, rejecting H_0 means the series IS stationary — that is good news. Always read the hypotheses carefully before interpreting a result.

P.6 — The t-test and the F-test

NEW CONCEPT

t-test for a single coefficient. Tests whether one regression coefficient (say β_1) is significantly different from zero. $H_0: \beta_1 = 0$ (X has no effect on Y). The t-statistic = (estimated β_1) / (standard error of β_1). If $|t| >$ critical value (roughly 2 for large samples at 5% significance), reject H_0 .

NEW CONCEPT

F-test (ANOVA). Tests whether the model as a whole is significant — i.e. whether at least one X variable has a non-zero effect on Y. H_0 : all β coefficients are zero. A significant F-test (small p-value) means the model explains a meaningful part of Y.

NEW CONCEPT

Standard error (SE). A measure of how precisely we have estimated a coefficient. A smaller SE means more precision. SE depends on the sample size and the variability of the data. When autocorrelation is present, the formula for SE becomes incorrect (see Section B.3), making t- and F-tests unreliable.

P.7 — The Chi-Squared (χ^2) Distribution

NEW CONCEPT

The chi-squared (χ^2) distribution is a probability distribution used in several statistical tests, including the LMSC autocorrelation test in this guide. Like the t-distribution, it has a parameter called degrees of freedom (df). You do not need to know how to compute χ^2 values by hand — the p-value from SPSS tells you directly whether to reject H_0 .

NEW CONCEPT

Degrees of freedom (df). Roughly speaking, df = the number of independent pieces of information used by a test. In the LMSC test $df = 1$ for a first-order test (df increases if you test for higher-order autocorrelation). In the Dickey-Fuller test, df is large (determined by sample size).

Overview — the time-series workflow

When you fit a regression model on time-series data, you go through the same four big steps every time. This guide is structured around them:

Recognise TS_A — Cautions	Diagnose TS_B — Autocorrelation	Stabilise TS_C — Stationarity
Time-series $\neq$ cross-section. Define lags X_{t-q} and differences ΔX_t . Beware too-few data.	Classical assumption A3 violated? Detect via LMSC test. Choose a dynamic model as solution.	Spurious regression risk. Use Dickey-Fuller. Detrend or difference if needed.
Model TS_D — Dynamic models		
Fit DL(q) or AR(1). Determine optimal lag length. Interpret the total multiplier.		

EXAM RECIPE

The exam steps. In an exam question on time-series regression you will be told which kind of model to fit (static, DL(q), or AR(1)). You then go through:

- Step 1 — Check stationarity of every variable: time plot + Dickey-Fuller. If non-stationary, take differences (or model a deterministic trend).
- Step 2 — Fit the model. For DL(q): determine the optimal lag length. Report coefficients and the total multiplier.
- Step 3 — Check assumptions, in particular autocorrelation with the LMSC test.

Big picture — connection with Part 1 (MR)

Everything you learned in multiple linear regression still applies: model specification, t-test, ANOVA, R^2 , the 5 classical assumptions, residual plots, multicollinearity, heteroskedasticity. The time-series part adds three issues on top:

- Autocorrelation — assumption A3 (uncorrelated errors) can fail because errors close in time are similar.
- Spurious regression / stationarity — variables that drift over time can look related even when they are not.
- Dynamic effects — past values of X (and sometimes Y) influence current Y. New models: DL(q), AR(1).

Cautions

Differences between cross-section and time-series data; lags and differences

Section	Topic	You should be able to...
A.1	Cross-section vs. time series	Explain how time-series data differ and why we must be careful.
A.2	Lagged variables X_{t-1} , X_{t-2} , ...	Build lags in SPSS; recognise LAG(x,1) notation; interpret.
A.3	Difference variables ΔX_t	Build differences in SPSS; recognise DIFF(x,1); interpret.
A.4	The three problems	Why we must be careful: data, autocorrelation, stationarity.

A.1 — Cross-section vs. time-series data

KEY POINT

Cross-section. Data on many economic units at one moment in time. The order of rows does not matter; observations are interchangeable. Notation X_i , with i the individual.

Time-series. Data on one economic unit collected over time. The order of rows carries information — shuffling destroys it. Notation X_t , with t the time index.

Panel data. Both at once — many units over many time periods (not exam material here, but worth keeping in mind).

Example — hourly wage

Cross-section: hourly wage for a sample of Belgian employees in 2010 — one observation per person.

Time-series: average hourly wage for the Belgian population in 2008, 2009, 2010, ... — one observation per year.

NEW CONCEPT

Why does the order matter? In a cross-section, observation 50 has nothing special to do with observation 49 — they are just two different people. In a time-series, today's value often depends on yesterday's value (e.g. today's inflation is close to last month's inflation). Ignoring this structure leads to wrong conclusions.

A.2 — Lagged variables

A lagged variable shifts the time index backwards. Notation:

FORMULA	$X_{t-1}, X_{t-2}, \dots, X_{t-q}$
NEW CONCEPT	What does 'lag' mean in practice? If t is quarter 3 of 1987, then X_{t-1} is the value of X in quarter 2 of 1987, and X_{t-2} is the value in quarter 1 of 1987. We 'look back' one or more periods.
INTERPRETATION	Example. If t is quarter 3 of 1987, then X_{t-1} is the value of X in quarter 2 of 1987, and X_{t-2} is the value in quarter 1 of 1987.
SPSS	Path: Transform → Create Time Series Function: Lag Order: 1, 2, 3, ... (one new variable per order) Internal name: LAG(inf,1), LAG(inf,2), ... for the variable inf.
WATCH OUT	You lose observations. For lag q you can compute the lag only from $t = q+1$ onwards, so the lagged variable has $n-q$ rather than n observations. This matters when n is small.

A.3 — Difference variables

A difference variable measures the change between two time points.

FORMULA	$\Delta X_t = X_t - X_{t-1}$ (absolute first difference) $\Delta_{12}X_t = X_t - X_{t-12}$ (difference of order 12 — useful for monthly data with yearly seasonality)
NEW CONCEPT	Why ' Δ '? The Greek capital letter Delta (Δ) is conventionally used in mathematics to denote 'change in'. So ΔX_t literally means 'the change in X at time t compared to the previous period'.
SPSS	Path: Transform → Create Time Series Function: Difference (absolute) — for the relative version use Function: Percentage change Internal name: DIFF(u,1), DIFF(u,2), ... for the variable u .

KEY POINT

Why differences matter:

- They convert an absolute level into a change, which is what most economic theories talk about (Δ unemployment, Δ inflation, ...).
- If you take the difference of a log-transformed variable you obtain a log-return: $\Delta \ln(X_t) = \ln(X_t / X_{t-1})$, often used in finance and macro.
- Differencing is the standard cure for a stochastic trend (see TS_C).

A.4 — Three problems with time-series regression

Whenever you do regression on time-series data, three issues need attention. The rest of this guide is structured around them.

PROBLEM 1 Data	PROBLEM 2 Autocorrelation	PROBLEM 3 Spurious / Stationarity
You need many observations spanning many time periods, and the data must be of comparable quality. Watch out for inflation when using monetary variables — use real (deflated) values.	Classical assumption A3 (uncorrelated errors) is often violated: an error today carries some memory of yesterday's error. → Chapter TS_B.	Two unrelated trending series can produce a hugely significant regression. The underlying issue is that the series are not stationary. → Chapter TS_C.

WATCH OUT

Sample size. Time-series analysis needs many observations. Rules of thumb:

- Prefer $n > 100$.
- Absolute minimum: $n \approx 35$ (and at least 10 observations per X-variable).
- Yearly data → you may need 35 years or more of history; consider monthly or quarterly data if available.
- For monetary variables, take inflation into account — use real (deflated) values, not nominal ones.

Autocorrelation

Errors that remember the past: detection with the LMSC test

Section	Topic	You should be able to...
B.1	Which assumption fails	Identify A3 (uncorrelated errors) and explain what fails.
B.2	First-order autocorrelation ρ	Define AR(1) for errors; interpret sign and magnitude of ρ .
B.3	Consequences	List effects on standard errors and (un)biasedness.
B.4	Detection — LMSC test	Carry out the LMSC test step by step with descriptive plots.
B.5	Solutions	Use a dynamic specification (DL or AR) to remove autocorrelation.

B.1 — Which classical assumption is violated?

Remember the assumptions under which the least-squares estimators are BLUE:

KEY POINT

Classical Assumptions Recap $Y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki} + \varepsilon_i$

- A1: $E(\varepsilon_i) = 0$
- A2: $\text{Var}(\varepsilon_i) = \sigma^2$ is constant (homoskedasticity)
- A3: $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$ for any pair ($i \neq j$)
- A4: X's are non-random and not exact linear combinations of each other
- A5: (optional) ε is normally distributed

NEW CONCEPT

What does A3 mean in plain English? It says that the error you make for observation i is completely unrelated to the error you make for observation j . For cross-sectional data this is often reasonable. For time-series data it is frequently violated: if your model underestimates Y in January, it is likely to also underestimate Y in February, because economic conditions change slowly.

KEY POINT

When A3 is violated — that is, when $\text{cov}(\varepsilon_i, \varepsilon_j) \neq 0$ for some $i \neq j$ — we say the errors are autocorrelated (or serially correlated).

In time-series data this is the rule rather than the exception: errors close in time tend to be similar.

B.2 — First-order autocorrelation

To model the relationship between two consecutive errors, we assume an AR(1) structure for the error term:

FORMULA

$$\varepsilon_t = \rho \varepsilon_{t-1} + u_t$$

ρ is the first-order autocorrelation coefficient; u is a classical (non-autocorrelated) error.

NEW CONCEPT

Reading this formula. The error today (ε_t) is partly a carry-over from yesterday's error ($\rho \times \varepsilon_{t-1}$) plus a new random shock (u_t). If $\rho = 0$, today's error is completely independent of yesterday's — that is A3 satisfied. If $\rho \neq 0$, there is a systematic pattern in the errors — A3 violated.

Interpreting the size of ρ

- $\rho = 0 \rightarrow$ no autocorrelation: $\varepsilon_t = u_t$ (classical error)
- We expect $-1 < \rho < 1$; otherwise the error term would explode.
- $|\rho|$ closer to 1 $\rightarrow$ the previous error matters more for predicting today's error $\rightarrow$ stronger autocorrelation.

$\rho > 0$ (positive)	$\rho < 0$ (negative)
If ε is high in one period, it tends to stay high in the next. The error series shows persistent runs above (or below) zero. Most common case in economics (shocks take time to fade).	The error oscillates — the sign tends to flip every period. Rare in time-series economics; sometimes seen in over-differenced data.

B.3 — Consequences

WATCH OUT

What goes wrong in OLS when A3 fails?

- The formulas for the standard errors are no longer correct, so confidence intervals and t-/F-tests become misleading.
- The OLS estimator is still linear and unbiased, but it is no longer best (no longer BLUE).

NEW CONCEPT

What does 'standard errors are wrong' mean in practice? Your t-statistic (coefficient / SE) becomes incorrect because the denominator (SE) is based on a formula that assumes A3. With autocorrelation, the SE is typically underestimated, making t-statistics appear larger than they really are. This can lead you to declare a relationship significant when it is not.

WATCH OUT

Big exception for AR(1) models. When the regression itself contains a lagged dependent variable (Y_{t-1} as regressor) and there is autocorrelation in the error term, the estimated coefficients become biased. This is far more serious than the general case.

INTERPRETATION

Practical takeaway.

- If autocorrelation is detected in a static or DL(q) model, your standard errors and tests are unreliable, but your coefficients are still telling you something true on average.
- If autocorrelation is detected in an AR(1) model, you can no longer trust the coefficient estimates themselves — this is a stronger reason to switch model specifications.

B.4 — Detection: the LMSC test

Step 0 — Graphical inspection

Plot residuals e_t against time. A clear pattern (waves, trends, clusters above/below zero) is a strong hint that autocorrelation is present.

The LMSC test — Lagrange Multiplier Serial Correlation

NEW CONCEPT

What is a Lagrange Multiplier (LM) test? It is a general method for testing whether a constraint (here: $\rho = 0$) is satisfied. We fit an 'auxiliary' regression that deliberately tries to find autocorrelation. If it finds a lot, we reject H_0 .

KEY POINT

Hypotheses:

- H_0 : no autocorrelation (equivalent to $\rho = 0$)
- H_1 : autocorrelation present (equivalent to $\rho \neq 0$)

FORMULA**Procedure (4 steps):**

1. Fit the original regression model and save residuals e_t .

2. Build the auxiliary regression — residuals on the original X's plus the lagged residual:

$$e_t = \alpha_0 + \alpha_1 X_{1t} + \dots + \alpha_k X_{kt} + \alpha_{k+1} e_{t-1} + u_t$$

3. Get the R^2 of the auxiliary regression and compute: $LM = n \cdot R^2$

4. Under H_0 , LM follows a χ^2 distribution with 1 degree of freedom.

Compare to a critical value or use the p-value.

WATCH OUT

- In the auxiliary model the sample size is $n - 1$ (you lose one observation because e_{t-1} has no value at $t = 1$).
- Higher-order test: add e_{t-2} , e_{t-3} , ... to the auxiliary regression. LM then follows a χ^2 with 2, 3, ... df.
- Reject $H_0 \rightarrow$ there IS autocorrelation. Do NOT reject $\rightarrow$ assumption A3 is plausibly satisfied.

Worked example — Phillips curve, quarterly Australian data

Original model: $Y_t = \beta_0 + \beta_1 X_t + \varepsilon_t$, with Y = inflation, X = change in unemployment rate, $n = 90$ (quarterly 1987Q1–2009Q3).

Auxiliary regression results	Computation
Sample size: $n = 90 - 1 = 89$ R^2 of the auxiliary model: 0.310211 Coefficient on e_{t-1} : 0.558 ($t = 6.219$, sig. .000)	$LM = n \cdot R^2 = 89 \times 0.310211 = 27.609$ p-value = $P(\chi^2(1) \geq 27.609) \approx 1.5 \times 10^{-7}$ Reject H_0 — autocorrelation is present.

Durbin-Watson — the alternative test

SPSS prints the Durbin-Watson statistic in the regression output. Values close to 2 suggest no autocorrelation; values close to 0 suggest strong positive autocorrelation; values close to 4 suggest strong negative autocorrelation.

WATCH OUT

Limitation. Durbin-Watson is not valid when the model contains a lagged dependent variable (i.e. for AR models). The LMSC test is preferred because it is valid in both static and AR models.

B.5 — Solutions

More advanced fixes exist (Generalized Least Squares, OLS with Newey-West / HAC standard errors, ...) but these are not exam material. For us, the standard route is:

KEY POINT

Re-specify the model to capture the dynamic structure of the data.

- Add lagged X's → a Distributed-Lag (DL) model. See TS_D.
- Add a lagged Y → an Autoregressive (AR) model. See TS_D.
- If the auxiliary regression keeps coming back significant, you almost certainly need a dynamic specification, not a different test.

INTERPRETATION

DL or AR — which one?

- If you suspect serial correlation, DL(q) is preferred over AR. Why? Because AR is biased when there's residual autocorrelation, whereas DL is only inefficient.
- DL is more prone to (quasi-)multicollinearity, but multicollinearity is a less severe issue than bias.
- In DL the standard hypothesis tests are not strictly valid either, but corrections exist.

Stationarity

Spurious regression, the generalized AR(1), and the Dickey-Fuller test

Section	Topic	You should be able to...
C.1	Spurious regression	Explain why two unrelated trending series can look correlated.
C.2	Definition of stationarity	State the formal (weak) stationarity conditions; first graphical check.
C.3	Generalized AR(1) — 4 cases	Discuss mean, variance and stationarity for each (α , λ , ρ) case.
C.4	Trends — deterministic vs. stochastic	Recognise both; choose detrending or differencing.
C.5	Solutions for non-stationarity	Apply detrending, model the trend, or take differences.
C.6	Dickey-Fuller test	Carry out the DF test (version 1 / version 2) with adapted critical values.

C.1 — Spurious regression

WATCH OUT

The problem. Two time series can show a strikingly significant regression even when there is no economic relationship at all. This is called spurious regression.

- Classic example: regress Belgian GDP per capita (1950–2004) on US population density — same period. R^2 is huge, the slope is significant, and yet the two have nothing to do with one another economically.
- The shared cause is that both series have a trend over time. The regression is picking up the joint drift, not a real relationship.

NEW CONCEPT

Why does trending cause this problem? If both X and Y are growing over time for completely unrelated reasons (e.g. population growth, productivity gains), a regression of Y on X will find a strong positive relationship simply because both happen to be going up together. This is misleading — the relationship is an artefact of time, not a real causal connection.

KEY POINT

The deeper reason spurious regression happens is that the variables are not stationary. Stationarity is therefore a new assumption added on top of the 5 classical MR assumptions whenever we work with time-series data.

C.2 — Definition of stationarity

Intuitively, a stationary time series does not trend or drift. If you cut it in half, the left half and the right half look roughly the same.

KEY POINT

Formal definition (weak stationarity). A time series Y_t is (weakly) stationary if for every t :

- $E(Y_t) = \mu$ (constant mean — series returns to its average)
- $\text{Var}(Y_t) = \sigma^2$ (constant variance)
- $\text{Cov}(Y_t, Y_{t-s}) = \gamma_s$ (the covariance depends only on the lag s , not on t)

NEW CONCEPT

Understanding the three conditions in plain English:

1. Constant mean: the series does not systematically move up or down over time. It fluctuates around a fixed level.
2. Constant variance: the series does not become more or less volatile over time. The spread of fluctuations stays roughly the same.
3. Covariance depends only on the gap: the relationship between values two periods apart is the same regardless of whether we look at 1950 or 2000.

First graphical check

Plot the series against time. Two warning signs of non-stationarity:

- The mean changes — there's a clear upward or downward trend, or seasonal pattern.
- The variance changes — the series gets noisier (or quieter) over time.

If the time plot looks like flat noise oscillating around a constant level, the series is plausibly stationary. For a formal verdict, use the Dickey-Fuller test (C.6).

C.3 — The generalized AR(1) model

In its most general form, the building block for studying stationarity is:

FORMULA

$$Y_t = a + \lambda t + \rho Y_{t-1} + \varepsilon_t$$

with ε_t white noise: independent, $N(0, \sigma^2)$, mean zero, no autocorrelation.

NEW CONCEPT

What is 'white noise'? White noise is the ideal error term — each observation is drawn independently at random with mean zero and constant variance. It contains no pattern whatsoever. When the error in a model is white noise, all the classical assumptions are satisfied.

Three parameters control the behaviour: the constant a , the trend coefficient λ , and the autoregressive coefficient ρ . The key question is whether $|\rho| < 1$ or $|\rho| = 1$.

Case 1 — Basic AR(1) $ \rho < 1, a = 0, \lambda = 0$	Case 2 — AR(1) + constant $ \rho < 1, a \neq 0, \lambda = 0$
Model: $Y_t = \rho Y_{t-1} + \varepsilon_t$ Behaviour: Returns to the mean 0. The further ρ is from 0, the longer the series stays on one side before reverting. ✓ STATIONARY	Model: $Y_t = a + \rho Y_{t-1} + \varepsilon_t$ Behaviour: Stationary about a non-zero mean $\mu = a / (1 - \rho)$. ✓ STATIONARY
Case 3 — AR(1) + linear trend $ \rho < 1, a \neq 0, \lambda \neq 0$	Case 4 — Random walk $ \rho = 1$
Model: $Y_t = a + \lambda t + \rho Y_{t-1} + \varepsilon_t$ Behaviour: Stationary about a deterministic trend $\mu + \delta t$. The series itself trends, but its deviation from the trend line is stationary. ✗ NOT STATIONARY	Model: $Y_t = a + Y_{t-1} + \varepsilon_t$ Behaviour: Stochastic trend. With $a = 0$: random walk; with $a \neq 0$: random walk with drift. Variance grows with t — definitely not stationary. ✗ NOT STATIONARY

KEY POINT

Summary card. AR(1) with $|\rho| < 1$ is the stationary family. AR(1) with $|\rho| = 1$ is the random-walk family — never stationary.

- $|\rho| < 1, \lambda = 0$: stationary (Cases 1–2). No problem in regression.
- $|\rho| < 1, \lambda \neq 0$: stationary about a deterministic trend (Case 3). Problem — fix with detrending.
- $|\rho| = 1$: stochastic trend / random walk (Case 4). Problem — fix with differencing.

C.3 (details) — What each case looks like

Case 1. $|\rho| < 1, a = 0, \lambda = 0 \rightarrow$ basic AR(1)

Model: $Y_t = \rho Y_{t-1} + \varepsilon_t$. By repeated substitution:

FORMULA

$$Y_t = \varepsilon_t + \rho \varepsilon_{t-1} + \rho^2 \varepsilon_{t-2} + \dots + \rho^t Y_0$$

$$E(Y_t) \approx 0 \text{ (constant)} \quad \text{Var}(Y_t) = \sigma^2 / (1 - \rho^2) \text{ (constant)}$$

$$\text{Cov}(Y_{t-s}, Y_t) = (\sigma^2 \rho^s) / (1 - \rho^2) \text{ (depends only on lag } s)$$

Correlation: $r(Y_{t-s}, Y_t) = \rho^s \rightarrow$ autocorrelation decays exponentially

INTERPRETATION

Example. $Y_t = 0.7 Y_{t-1} + \varepsilon_t$. Because $\rho = 0.7 > 0$, the series tends to stay a while above (or below) its mean before reverting. The closer ρ gets to 1, the slower the revert.

Case 2. $|\rho| < 1, a \neq 0, \lambda = 0 \rightarrow \text{AR}(1)$ with non-zero mean

Model: $Y_t = a + \rho Y_{t-1} + \varepsilon_t$.

FORMULA

$$E(Y_t) \approx \mu = a / (1 - \rho)$$

$$\text{Var}(Y_t) = \sigma^2 / (1 - \rho^2)$$

$$\text{Cov}(Y_{t-s}, Y_t) = (\sigma^2 \rho^s) / (1 - \rho^2)$$

All three are constant in $t \rightarrow$ the series is stationary about a non-zero mean μ .

Case 3. $|\rho| < 1, a \neq 0, \lambda \neq 0 \rightarrow$ stationary about a deterministic trend

Model: $Y_t = a + \lambda t + \rho Y_{t-1} + \varepsilon_t$.

FORMULA

$$\delta = \lambda / (1 - \rho) \quad \mu = (a - \rho\delta) / (1 - \rho)$$

$Y_t - \mu - \delta t$ is a stationary AR(1) about 0.

INTERPRETATION

Example. $Y_t = 1 + 0.05t + 0.7 Y_{t-1} + \varepsilon_t$.

- $\delta = 0.05 / (1 - 0.7) = 1/6$
- $\mu = (1 - 0.7 \times 1/6) / (1 - 0.7) = 53/18$
- So $Y_t - 53/18 - (1/6)t$ is a stationary AR(1) about 0.

Case 4. $|\rho| = 1 \rightarrow$ random walk**KEY POINT**

1. Random walk ($a = 0, \lambda = 0$): $Y_t = Y_{t-1} + \varepsilon_t$

$E(Y_t) = y_0$ (constant), but $\text{Var}(Y_t) = t\sigma^2 \rightarrow$ variance grows with time. ✗

2. Random walk with drift ($a \neq 0, \lambda = 0$): $Y_t = a + Y_{t-1} + \varepsilon_t$

$E(Y_t) = ta + y_0 \rightarrow$ mean drifts up (or down) over time. ✗

3. Random walk with drift and trend ($a \neq 0, \lambda \neq 0$): $Y_t = a + \lambda t + Y_{t-1} + \varepsilon_t$

Mean grows like $ta + \frac{1}{2}t(t+1)\lambda \rightarrow$ even faster drift. ✗

C.4 — Deterministic vs. stochastic trends

Deterministic trend

Stochastic trend

Comes from a λt term $\rightarrow$ mean changes, variance stays constant. Types: linear ($\mu + \delta t$); seasonal (monthly/quarterly dummies). Example: monthly unemployment figures in Flanders — same months of the year cluster together.

Comes from $|\rho| = 1 \rightarrow$ random walk. Mean may or may not drift, but variance grows with time. Types: random walk; random walk with drift; random walk with drift and trend. Example: many financial prices behave roughly as random walks with drift.

WATCH OUT

In practice it is often hard to tell which is which just from a plot. The standard pragmatic answer: include both possibilities (model 3 of the DF test) when you are unsure — i.e. random walk with drift and trend.

C.5 — Solutions for non-stationarity

If the problem is a deterministic trend ($\lambda \neq 0$, $|\rho| < 1$)

KEY POINT

Option 1 — detrending. Replace the non-stationary Y_t by $Y_t - \mu - \delta t$, which is stationary.

Option 2 — model the trend. Keep Y_t but add a linear-trend term to the model:

$$Y_t = \beta_0 + \beta_1 X_t + \varepsilon_t \text{ becomes } Y_t = \alpha + \lambda t + \beta_1 X_t + \varepsilon_t$$

Option 3 — seasonal differencing. For seasonal trends (e.g. monthly data), use $\Delta_{12} X_t = X_t - X_{t-12}$ — the change vs. the same month one year earlier.

If the problem is a stochastic trend ($|\rho| = 1$)

KEY POINT

Take first differences. Detrending does NOT work here — it is for deterministic trends only. Instead, replace each variable by its first difference:

$$\Delta Y_t = Y_t - Y_{t-1}, \Delta X_t = X_t - X_{t-1}$$

- If after one round of differencing the variables are stationary, fit your regression on the differences (not the levels).

- Why it works: if $Y_t = Y_{t-1} + \varepsilon_t$, then $\Delta Y_t = \varepsilon_t$ — white noise, stationary.

- If even the first differences are not stationary, try second differences $\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1}$.

INTERPRETATION

Worked example — GDP and Gross National Income (Belgium, 1950–2004).

- Both series clearly trend upward over time. A regression in levels would be spurious.
- After taking first differences (or relative differences), the trends disappear. The relationship between ΔGDP and ΔGNI remains and is now economically meaningful.

C.6 — The Dickey-Fuller test

Until now, choosing between stationary and non-stationary was a judgement call from a time plot. The Dickey-Fuller (DF) test makes it formal: it is a unit-root test — it tests whether $\rho = 1$ in the generalized AR(1).

NEW CONCEPT

What is a 'unit root'? The term comes from the mathematics of the AR(1) model. When $\rho = 1$, the model has a 'unit root' — the process does not revert to a mean and the variance grows over time. Testing for a unit root is the same as testing whether the series is a random walk (non-stationary).

KEY POINT

Hypotheses (re-parameterised).

- $H_0: \rho = 1$ (unit root — series is NOT stationary)
- $H_1: |\rho| < 1$ (series is stationary, possibly about a deterministic trend)

FORMULA

The trick: take first differences and reparameterise. Setting $\alpha_1 = \rho - 1$:

Version 1 — no trend: $\Delta Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \varepsilon_t$

Version 2 — with linear trend: $\Delta Y_t = \alpha_0 + \lambda t + \alpha_1 Y_{t-1} + \varepsilon_t$

Reject $H_0 \iff \alpha_1 < 0$ (significantly).

WATCH OUT

Adapted critical values. Under H_0 the t-statistic is NOT t-distributed — you cannot use the usual t-critical values. Use the table below.

DF version	1%-level	5%-level	10%-level
Version 1 — without trend	−3.43	−2.86	−2.57
Version 2 — with trend	−3.96	−3.41	−3.13

Decision rule. If the t-stat on Y_{t-1} is more negative than the critical value, reject $H_0 \rightarrow$ the series is stationary.

KEY POINT

Procedure (4 steps).

1. Plot the data versus time.
2. No visible trend? Use Version 1 (no trend). Clear trend? Use Version 2 (with trend). Unsure? Use Version 2 — it's the safer default.
3. Run the chosen DF regression with ΔY_t as response and Y_{t-1} as regressor (plus the time variable for Version 2).
4. Read the t-statistic for the coefficient of Y_{t-1} and compare to the adapted critical values above.

Reference — mean of a stationary AR(1)

Model	Mean / behaviour
$Y_t = \rho Y_{t-1} + \varepsilon_t$	Stationary, mean 0
$Y_t = a + \rho Y_{t-1} + \varepsilon_t$	Stationary, mean $\mu = a / (1 - \rho)$
$Y_t = a + \lambda t + \rho Y_{t-1} + \varepsilon_t$	Stationary about a deterministic trend $\mu + \delta t$, with $\delta = \lambda / (1 - \rho)$, $\mu = (a - \rho\delta) / (1 - \rho)$

Dynamic models

DL(q), AR(1), total multipliers, and the time-series exam workflow

Section	Topic	You should be able to...
D.1	Distributed lag model DL(q)	Write the equation; interpret coefficients; find total multiplier.
D.2	Optimal lag length	Pick q starting from q_max by removing non-significant top lags.
D.3	Autoregressive model AR(1)	Write the equation; interpret coefficients; find total multiplier $\beta/(1-\gamma)$.
D.4	Link $DL(\infty) \leftrightarrow AR(1)$	Show how AR(1) is an infinite-lag DL; derive geometric weights.
D.5	Three-step exam workflow	Combine stationarity, fit, assumption check on a real exam problem.

D.1 — Distributed lag model DL(q)

In a static model, Y depends only on the current X. But economic effects are rarely instantaneous: training today changes accident counts months later; a tax cut today affects consumption next quarter. To capture this we use a distributed lag model:

FORMULA

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \dots + \beta_q X_{t-q} + \varepsilon_t$$

$$\text{Total multiplier} = \sum \beta_k \text{ (sum over } k = 0, 1, \dots, q)$$

NEW CONCEPT

Reading the DL(q) model. Each β coefficient captures the effect of X at a different delay. β_0 is the immediate effect; β_1 is the effect felt one period later; β_q is the effect felt q periods later. Adding them all gives the total multiplier — the full long-run effect of a permanent change in X.

SPSS

How to set up DL(q) in SPSS:

- Transform → Create Time Series → Function: Lag, Order: 1, 2, ..., q (to make the lagged variables)
- Analyze → Regression → Linear: put Y as Dependent and $X_t, X_{t-1}, \dots, X_{t-q}$ as Independents.
- Sample size shrinks to $n - q$.

Interpreting the coefficients of DL(q)

KEY POINT

As in any regression: β_k is the average change in Y when X_{t-k} rises by 1 unit, holding all other variables constant.

Temporary change (X rises by 1, then returns)	Maintained change (X rises by 1 and stays high)
Immediate effect: β_0 After 1 period: β_1 After 2 periods: β_2 , and so on The effect dies out once you go beyond lag q .	Immediate effect: β_0 After 1 period: $\beta_0 + \beta_1$ After 2 periods: $\beta_0 + \beta_1 + \beta_2$ After q or more periods the total effect is $\sum \beta_k = \text{total multiplier}$

INTERPRETATION

In words.

- The total multiplier is the long-run effect on Y of a permanent 1-unit rise in X .
- It is the answer to the exam-style question: 'by how much will Y eventually go up if X increases by 1 unit and stays there forever?'

D.2 — Choosing the optimal lag length q

KEY POINT

Procedure (top-down). Start from a maximum lag $q_{\max}$ (given in the exam, e.g. $q_{\max} = 3$ or 4) and shrink it.

1. Fit the $DL(q_{\max})$ model. Look at the t-test for the highest-order lag, $\beta_{q_{\max}}$.
2. If it is not significant, drop that lag and refit $DL(q_{\max}-1)$. Repeat.
3. Stop as soon as the t-test for the largest remaining lag is significant. That q is your optimal lag length.

INTERPRETATION

Worked example — safety training and accident losses.

- Goal: model monthly accident losses (Y , euro) as a function of monthly safety-training hours (X), with $q_{\max} = 4$.
- Build X_{t-1} , X_{t-2} , X_{t-3} , X_{t-4} and fit $DL(4)$.
- Suppose the t-test on β_4 is not significant — drop the lag, refit $DL(3)$; β_3 also not significant — refit $DL(2)$; β_2 not significant — refit $DL(1)$; β_1 is significant. Stop: optimal lag is $q = 1$.
- Now read off β_0 and β_1 , compute the total multiplier, and interpret it.

D.3 — Autoregressive model AR(1)

In a $DL(q)$ model the dynamics come from past X 's. We can instead let past Y carry the memory:

FORMULA

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$$

Total multiplier = $\beta / (1 - \gamma)$ (geometric-series formula; requires $|\gamma| < 1$)

NEW CONCEPT

Why does Y_{t-1} appear? Many economic variables carry their own history — GDP this quarter depends partly on GDP last quarter. Including Y_{t-1} as a regressor captures this inertia parsimoniously. Instead of needing many lag coefficients (like DL), a single γ coefficient does the work.

Interpreting AR(1)

- Immediate effect (temporary or maintained 1-unit rise in X): β .
- Long-run effect of a maintained 1-unit rise in X: $\beta / (1 - \gamma) \rightarrow$ the total multiplier.
- A temporary 1-unit rise in X gives a one-off jump of β , then a smaller residual effect on Y in every subsequent period that fades by a factor γ .

INTERPRETATION

Worked example — $Y = 20 + 10X + 0.5 Y_{t-1}$. In year 5, X temporarily rises by 1.

- Year 5: Y jumps by 10 (the β coefficient).
- Year 6: Y rises by $0.5 \times 10 = 5$ relative to its original level — even though X is back to normal — because Y_5 was elevated.
- Year 7: effect is $0.5^2 \times 10 = 2.5$; year 8: $0.5^3 \times 10 = 1.25$, ...
- Total multiplier = $10 / (1 - 0.5) = 20$. If X had stayed at the new level, Y would eventually rise by 20.

INTERPRETATION

Worked example — education spending and GDP growth.

Yearly US data since 1910. X = dollars invested in education per child, Y = GDP growth (%).

Fitted AR(1): $\hat{Y}_t = 1.01 + 0.009 X_t + 0.627 Y_{t-1}$

- Total multiplier = $\beta / (1 - \gamma) = 0.009 / (1 - 0.627) = 0.024$.
- Interpretation: if the investment per child increases by 1 dollar and is maintained over the years, then GDP growth rises on average by 0.024 percentage points in the long run.

D.4 — Link between AR(1) and DL(∞)

Why does the AR(1) total multiplier have that specific shape $\beta / (1 - \gamma)$? Because AR(1) is secretly a DL model with infinitely many lags.

FORMULA	$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$ $= \alpha + \beta X_t + \gamma(\alpha + \beta X_{t-1} + \gamma Y_{t-2}) + \varepsilon_t$... repeating to infinity ... $Y_t = \alpha' + \beta X_t + \beta\gamma X_{t-1} + \beta\gamma^2 X_{t-2} + \beta\gamma^3 X_{t-3} + \dots + \varepsilon'_t$
KEY POINT	Geometric-series consequence. The total multiplier of the AR(1) model is: $\Sigma \beta_k = \beta(1 + \gamma + \gamma^2 + \gamma^3 + \dots) = \beta / (1 - \gamma)$ This is exactly the geometric series $1 + q + q^2 + \dots = 1 / (1 - q)$, converging when $\gamma < 1$.
NEW CONCEPT	Geometric series refresher. If $q < 1$, then $1 + q + q^2 + q^3 + \dots = 1 / (1 - q)$. Example: $1 + 0.5 + 0.25 + 0.125 + \dots = 1 / (1 - 0.5) = 2$. The series converges because each term is a fixed fraction of the previous one, eventually becoming negligible.
INTERPRETATION	Why does this matter? • AR(1) gives you long memory with only two coefficients (β and γ), instead of many β_k's. Much less prone to (quasi-)multicollinearity than DL(q) with large q. • Drawback: AR(1) is biased when there is residual autocorrelation (see B.3). DL(q) is the safer choice when you suspect that.

D.5 — The three-step exam workflow

This is the procedure to follow whenever an exam question asks you to fit a static, DL(q), or AR(1) model on time-series data. The model type will be specified — you do not have to choose. You go through three steps in order.

STEP 1 Stationarity check	STEP 2 Fit the model	STEP 3 Check assumptions
• Time plot of every variable. • Choose DF version (no-trend or with-trend). • Run DF on each variable, read t-stat, compare with adapted critical values. • Unit root anywhere? → take Δ of all variables, re-test on Δ.	• Static / DL(q) / AR(1) — given in the question. • For DL: find optimal q starting from q_max (drop top lags until significant). • Report coefficients + total multiplier ($\Sigma \beta$ or $\beta/(1-\gamma)$). • Interpret with the right words: temporary vs. maintained, log-return vs. unit change.	• State H_0 / H_1 in plain English. • Write the auxiliary regression (X's + lagged residual). • Compute LM = $n \cdot R^2$; compare to $\chi^2(1)$. • Conclude — autocorrelation present or assumption satisfied. • Also briefly mention the other classical assumptions.

Step-by-step worked example — consumption ~ income (log-log)

INTERPRETATION

Setup. Both income (X) and consumption (Y) are log-transformed, then we fit a DL(q) model on their differences.

STEP 1 — Stationarity.

- Time plots of Y and X show clear upward trends. Run DF on the levels — you fail to reject H_0 for both. Both variables have a unit root.
- Take first differences (which are log-returns): ΔY and ΔX . Run DF Version 1 (no visible trend in the differences).
- ΔY : $t = -9.95$, well below -2.86 . Reject $H_0 \rightarrow$ stationary.
- ΔX : $t = -12.72$, also well below -2.86 . Reject $H_0 \rightarrow$ stationary.
- Both variables are stationary after differencing; we proceed with the differences.

STEP 2 — Fit the model.

- Build lags of ΔX up to $q_{\max} = 3$.
- DL(3): β_3 not significant $\rightarrow$ drop. DL(2): β_2 not significant $\rightarrow$ drop. DL(1): β_1 is significant. Stop. Optimal lag is $q = 1$.
- Fitted model: $\Delta \hat{Y}_t = 0.004 + 0.35 \Delta X_t + 0.182 \Delta X_{t-1}$
- Total multiplier = $0.35 + 0.182 = 0.532$.
- Interpretation (log-returns!): if the relative change in income rises by 1 percentage point and is maintained, then the relative change in consumption rises on average by 0.532 percentage points after one quarter.

STEP 3 — Assumption check (LMSC test).

- Auxiliary regression: $e_t = \alpha_0 + \alpha_1 \Delta X_t + \alpha_2 \Delta X_{t-1} + \alpha_3 e_{t-1} + u_t$.
- Sample size $n = 161$ (one observation lost to the lagged residual).
- R^2 of the auxiliary regression: 0.002618. $LM = 161 \times 0.002618 = 0.4215$.
- $p\text{-value} = P(\chi^2(1) \geq 0.4215) = 0.5162$. Do not reject $H_0 \rightarrow$ no significant autocorrelation. Assumption A3 is plausibly satisfied. ✓

EXAM-DAY CHECKLIST

Time-Series Regression

STEP 1 STATIONARITY TS_C	STEP 2 FIT MODEL TS_D	STEP 3 ASSUMPTIONS TS_B
• Time plot of every variable. • Choose DF version (no-trend or with-trend). • Run DF on each variable, read t-stat, compare with adapted critical values. • Unit root anywhere? → take Δ of all variables, re-test on Δ.	• Static / DL(q) / AR(1) — given in the question. • For DL: find optimal q starting from q_max (drop top lags until significant). • Report coefficients + total multiplier ($\Sigma\beta$ or $\beta/(1-\gamma)$). • Interpret with the right words: temporary vs. maintained, log-return vs. unit change.	• State H_0 / H_1 in plain English. • Write the auxiliary regression (X's + lagged residual). • Compute LM = $n \cdot R^2$; compare to $\chi^2(1)$. • Conclude — autocorrelation present or assumption satisfied.

Formula sheet — the four formulas you really need

FORMULA

1. DL(q) model and its total multiplier:

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \dots + \beta_q X_{t-q} + \varepsilon_t ; \Sigma\beta_k$$

2. AR(1) model and its total multiplier:

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t ; \beta / (1 - \gamma)$$

3. LMSC test — auxiliary regression and statistic:

$$e_t = \alpha_0 + \alpha_1 X_{1t} + \dots + \alpha_k X_{kt} + \alpha_{k+1} e_{t-1} + u_t$$

$$LM = n \cdot R^2 \sim \chi^2(1)$$

4. Dickey-Fuller test — two versions:

$$\text{Version 1: } \Delta Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \varepsilon_t$$

$$\text{Version 2: } \Delta Y_t = \alpha_0 + \lambda t + \alpha_1 Y_{t-1} + \varepsilon_t$$

Critical values: 5% → -2.86 (V1), -3.41 (V2). Reject H_0 if t-stat is more negative.

Common exam pitfalls

WATCH OUT

1. 'Reject H_0 means stationary.' For DF, yes — Reject = Stationary. But many people confuse DF with the LMSC test, where Reject = autocorrelation present (i.e. bad news). Always re-read the hypotheses.
2. 'Take differences whenever in doubt.' Over-differencing creates artificial negative autocorrelation. Only difference if the DF test (or the time plot) tells you to.
3. 'Use Durbin-Watson for AR(1).' Durbin-Watson is not valid when a lagged dependent variable is in the model. Use LMSC.
4. 'Total multiplier = β_0 only.' No — for DL(q) it is the sum of all β_k 's. For AR(1) it is $\beta / (1 - \gamma)$, not just β .
5. 'Interpret as units when X is logged.' If you took log-differences (log-returns), interpret as percentage-point changes, not unit changes.
6. 'Skip the assumption check.' The LMSC test is worth easy marks and must be reported with H_0/H_1 , LM-statistic, p-value, and a conclusion sentence.
7. 'Use the t-distribution for the DF t-stat.' Under H_0 it is not t-distributed. Always use the adapted critical values ($-2.86, -3.41, \dots$).

THE WHOLE GUIDE IN FOUR LINES

Recognise the data is time series → define lags, define differences.

Diagnose autocorrelation with LMSC — but think first about whether you need a dynamic model.

Stabilise with detrending (deterministic trend) or differencing (stochastic trend / unit root).

Model with DL(q) or AR(1) — find the optimal q, interpret the total multiplier, run LMSC.